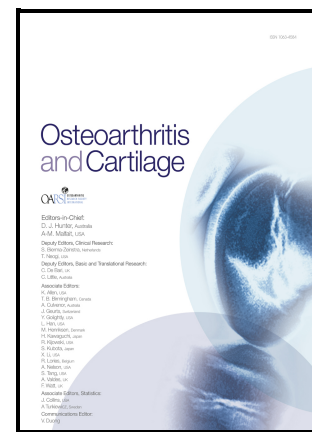


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Association between denosumab use and risk of osteoarthritis among adults with osteoporosis in a real-world cohort

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Abstract

Objective: To estimate the effect of denosumab compared with bisphosphonates on reducing the risk of knee or hip osteoarthritis (OA) among adults with osteoporosis in a real-world cohort.

Method: This new-user, active-comparator retrospective real-world cohort utilized electronic health records from the TriNetX Network. Participants diagnosed with osteoporosis between January 1, 2014 and December 31, 2023 were selected. We used propensity-score matching (PSM) to balance potential confounders between denosumab users and bisphosphonate users. Cox proportional hazards models were used to estimate the hazard ratios (HRs) and 95% confidence intervals (CIs) of incident knee and hip OA by comparing PSM groups.

Results: 59,157 new users of denosumab were matched on propensity score to 59,157 participants of bisphosphonate users (mean age 70 years; 87% female). The 10-year incidence probability of knee and hip OA in denosumab users was 19.2% (4,839 events) and 14.5% (3,396 events), while in bisphosphonate users it was 21.4% (5,389 events) and 14.4% (3,388 events). Compared with bisphosphonate, denosumab users were significantly associated with a reduced risk of any OA (HR 0.96, 95% CI 0.94-0.98), knee OA (HR, 0.87, 95% CI, 0.84-0.91), but not with hip OA (HR, 0.98, 95% CI, 0.94-1.03), or thumb carpometacarpal OA (HR 0.94, 95% CI 0.87-1.01). These associations were more pronounced among older participants (≥ 65 years, HR 0.88, 95% CI 0.84-0.91 for knee OA), females (HR 0.87, 95% CI 0.83-0.91 for knee OA), and individuals of Asian ethnicity (HR 0.73, 95% CI 0.63-0.84 for knee OA).

Conclusions: Denosumab use was associated with a reduced risk of incident knee OA compared with bisphosphonates in adults with osteoporosis, with the association being more pronounced among older adults, females, and individuals of Asian ethnicity. This study suggests that denosumab could attenuate the development of knee OA, warranting further clinical trials.

Keywords: Osteoarthritis; osteoporosis; denosumab; bisphosphonate

Introduction

Osteoarthritis (OA) is a highly prevalent chronic joint disease affecting over 595 million individuals worldwide, with an estimated global prevalence of 22% in individuals aged 40 and above.¹ It is a leading cause of chronic pain and disability, accounting for more than 7% of all years lived with disability globally.² Despite its widespread prevalence and significant disease burden on healthcare systems, there are currently no disease-modifying drugs approved for OA treatment, leaving a critical gap in therapeutic options.³ Current interventions primarily focus on symptomatic relief and function improvement, such as pain management and joint replacement surgery, underscoring the urgent need for novel therapeutic or preventive strategies that address the underlying pathophysiological mechanisms of OA.⁴

Denosumab, a humanized monoclonal antibody targeting the receptor activator of nuclear factor κ B ligand (RANKL), is a potent antiresorptive agent widely used in managing osteoporosis.⁵ By inhibiting RANKL signaling, denosumab suppresses osteoclastogenesis, reduces bone resorption, and promotes subchondral bone remodeling.⁶ Emerging preclinical evidence suggests that RANKL downregulation may also mitigate oxidative stress-induced apoptosis and cartilage degradation, which are key processes implicated in OA pathogenesis.⁷ Notably, a randomized controlled trial demonstrated that denosumab exerted structure-modifying effects in erosive hand OA, improving subchondral bone integrity and preventing joint erosion.⁸ However, evidence on the impact of denosumab on large joint OA, particularly in the knee and hip, remains sparse.

The paucity of real-world evidence examining denosumab's potential to mitigate the risk of OA in weight-bearing joints represents a critical knowledge gap. Therefore, this study aimed to evaluate the association

between denosumab use and the risk of incident knee and hip OA in a real-world cohort of adults with osteoporosis. By leveraging electronic health records and propensity-score matching (PSM), this investigation provides valuable insights into denosumab's potential as a preventive agent for OA and informs the design of future clinical trials targeting this debilitating condition.

Methods

Data source and study design

This retrospective cohort study utilized data from the TriNetX US Collaborative Network to compare the risk of OA among adults with osteoporosis treated with either denosumab or bisphosphonates. The dataset was extracted on September 10, 2024. TriNetX is a global federated health research network providing access to de-identified electronic health records (EHRs), including diagnoses, procedures, medications, laboratory values, and genomic information data, contributed by 66 large healthcare organizations across the United States.

The majority of participating institutions are academic medical centers with both inpatient and outpatient services spanning all 50 U.S. states. TriNetX adheres to the Health Insurance Portability and Accountability Act (HIPAA), U.S. and other applicable data privacy regulations. TriNetX is ISO 27001:2013 certified and maintains an Information Security Management System (ISMS) to ensure the protection of healthcare data and compliance with the HIPAA Security Rule.

Data within TriNetX were mapped to standardized clinical terminologies (e.g., SNOMED-CT to ICD-10, NDC to RxNorm) and transformed into a proprietary schema following rigorous data quality assessments⁹. All study variables were binary, categorical (which is expanded to a set of binary columns), or continuous. Age was guaranteed to exist. Missing sex values were represented using 'unknown sex'. The missing data for ethnic grouping and ethnicity were presented as 'unknown race' or 'unknown ethnicity'. For other variables, including medical conditions, procedures, laboratory tests, and socioeconomic determinant health, the value was either present or absent, so 'missing' is not pertinent.

All data available on the TriNetX platform was de-identified in accordance with Section §164.514(a) of the HIPAA Privacy Rule, and the de-identification process was validated by a qualified expert as defined in Section §164.514(b)(1). As the study was based solely on de-identified data, it was exempt from Institutional Review Board (IRB) review. Nonetheless, the use of TriNetX for this research was approved by the IRB of Chung Shan Medical University Hospital (approval number: CS2-21176).

Study population

We identified adults aged ≥ 45 years who had at least two diagnoses of osteoporosis (ICD-10: M80 and M81) between January 1, 2014, and December 31, 2023, within the TriNetX US Network. To ensure a new user design, we excluded individuals who had received any prescriptions for denosumab or bisphosphonates before December 31, 2013. Eligible participants had newly initiated either denosumab or bisphosphonate treatment during the study period, with an osteoporosis diagnosis recorded within one week before or after the prescription date. The date of the first prescription for denosumab or bisphosphonates was defined as the index date. A total of 360,283 participants met the inclusion criteria.

Denosumab group

We identified 104,340 individuals with a diagnosis of osteoporosis who subsequently initiated denosumab treatment. To minimize confounding, we excluded participants with advanced chronic kidney disease (defined as an estimated glomerular filtration rate [eGFR] < 30 mL/min/1.73 m², dialysis, or diagnosis of end-stage renal disease (ESRD) with ICD-10 codes N18.4–N18.6), Paget's disease of bone, or a prior diagnosis of OA before denosumab initiation. After applying these criteria, a total of 59,157 participants were included in the final analysis for the denosumab group (Figure 1).

Bisphosphonate group

We identified 255,943 individuals with a diagnosis of osteoporosis who subsequently initiated bisphosphonate treatment, either as an intravenous injection or oral administration. To maintain bisphosphonate treatment group consistency, individuals who switched to denosumab before December 31, 2023, were excluded. We also applied the same exclusion criteria, including advanced chronic kidney disease, Paget's disease of bone, and prior diagnosis of OA before bisphosphonate treatment initiation. After exclusions, 161,044 participants

remained in the bisphosphonate group for analysis (Figure 1). To balance exposure specificity and representativeness, we excluded bisphosphonate users who subsequently received denosumab to prevent contamination of the comparator group, while retaining denosumab initiators regardless of subsequent therapy under an intention-to-treat (ITT) approach. This design ensures that the observed differences primarily reflect outcomes associated with the initial treatment choice rather than subsequent medication changes.

Outcome and follow-up

The primary outcome was the new onset of OA (Supplementary Table 1), specifically affecting the knee, hip, or thumb carpometacarpal. Incident OA was identified using ICD-10-CM codes M15 (polyarthrosis), M16 (hip OA), M17 (knee OA), and M18 (first carpometacarpal joint OA) ¹⁰. The study assessed the 10-year risk of newly diagnosed OA (ICD-10-CM: M15–M19) in the denosumab and bisphosphonate groups. For each participant, follow-up began at the index date, defined as the date of the first prescription of denosumab or bisphosphonate, and continued until the earliest occurrence of the following: a new diagnosis of OA, or the last recorded encounter within the 10-year observation period. The median follow-up time before PSM was 1,297 days in the denosumab group and 1,370 days in the bisphosphonate group. After PSM, the median follow-up time was 1,298 days for the denosumab group and 1,252 days for the bisphosphonate group.

Co-variables

To control for potential confounding factors, we included a comprehensive set of covariates assessed either on the index date or within the six months preceding it. These covariates encompassed demographic characteristics (age, sex, ethnicity, race, and socioeconomic status), lifestyle-related factors such as body mass index (BMI), nicotine dependence (ICD-10: F17) as a proxy for smoking, and alcohol-related disorders (ICD-10: F10) as a proxy for alcohol consumption. Healthcare encounters during the baseline period, including preventive care visits, hospital admissions, and emergency department presentations within the six months preceding the index date, were also identified. We also accounted for a broad range of comorbid conditions, including hypertensive diseases (ICD-10: I10-I1A), ischemic heart diseases (ICD-10: I20-I25), heart failure (ICD-10: I50), diabetes mellitus (ICD-10: E08-E13), dyslipidemia (E78), acute kidney failure (ICD-10: N17), chronic kidney disease (CKD, ICD-10: N18), diseases of liver (ICD-10: K70-K77), spondylopathies (M45-M49), inflammatory polyarthropathies (ICD-10: M05-M14), depressive episode (F32), chronic lower

respiratory diseases (ICD-10: J40-J4A), gout (ICD-10: M10), neoplasms (ICD-10: C00-D49), and malignant neoplasm of bone (ICD-10: C41). Co-medications considered in the analysis included systemic corticosteroids (ATC code: H02), non-steroidal anti-inflammatory drugs (NSAIDs; ATC code: M01A), disease-modifying antirheumatic drugs, opioids (ATC code: N02A), estrogens (ATC code: G03C), and selective estrogen receptor modulators (SERMs; ATC code: G03XC).

Statistical analysis

To address baseline differences between treatment groups, we performed PSM based on key covariates, including demographics, lifestyle factors, healthcare encounters, co-morbidities, and co-medication. A propensity score for each participant was estimated using logistic regression implemented through the validated machine learning library scikit-learn in Python. These scores, ranging from 0 to 1, represented the predicted probability of receiving denosumab treatment given the individual's covariate profile.

TriNetX aggregates data from a global federated health network. Each participating healthcare organization contributed a covariate matrix for its patients, which was subsequently pooled for analysis. To minimize selection bias, we employed greedy nearest neighbour matching with a caliper of 0.1 pooled standard deviation, and the order of records was randomized to prevent order-dependent bias in the matching process. After PSM, there were 59,157 pairs of denosumab users and bisphosphonate users selected for analysis (Figure 1). The TriNetX Live Analytics environment does not currently support inverse probability of treatment weighting (IPTW) for risk estimation. Therefore, PSM was selected as the primary method for causal adjustment (Supplementary Figure 1).

Time-to-event analysis was performed using Kaplan–Meier methods to estimate the cumulative incidence of OA over daily intervals. Censoring was applied at the last recorded clinical encounter within the defined time window to account for loss to follow-up. Cox proportional hazards models were used to compare the matched cohorts, with the proportional hazards assumption assessed via the generalized Schoenfeld residuals approach. Hazard ratios (HRs) and corresponding 95% confidence intervals (CIs) were estimated using the survival package (v3.2-3) in R, embedded within the TriNetX Analytics Platform.

We performed stratified analyses to examine the heterogeneity of OA risk in denosumab users compared with bisphosphonate users across different subgroups categorized by sex, race (White, Black or African American, and Asian), age (45-65 years versus 65 years or older), and BMI (<25, 25-30, and ≥ 30). To assess the robustness of the primary analysis, we performed several sensitivity analyses. First, to address concerns about differential administration reliability, we conducted separate comparisons of denosumab with injectable and oral bisphosphonates. Second, we included separating injectable and oral forms of denosumab and applying varying lag periods (3, 6, and 12 months) for the initiation of follow-up. All statistical analyses were two-sided, with significance defined as $P < 0.05$.

Results

Characteristics of study cohorts

After PSM, 59,157 denosumab users were successfully matched 1:1 with 59,157 bisphosphonate users (Figure 1). Table 1 describes the baseline characteristics of participants before and after PSM. Before PSM, those who used denosumab were older (69.6 vs 67.5 years), less likely to be Hispanic or Latino ethnic group (3.2% vs 5.7%), and less likely to be Black or African American (3.2% vs 6.2%), compared with bisphosphonate users. Denosumab users also had lower eGFR, lower rates of using preventive medicine services, lower prevalence of baseline hypertensive diseases and dyslipidemia, but had a higher prevalence of neoplasms. In contrast, the two groups were similar in sex, BMI, socioeconomic status, lifestyle, co-medication like NSAIDs use, opioids use, and corticosteroids use, and most comorbidities like diabetes, ischemic heart disease, and gout. In the matched populations, baseline characteristics of the two groups were all comparable, with the standardized mean difference of less than 0.1 (Table 1).

Main analysis

In the matched cohorts, the 10-year incidence of any OA was 50.8% (15,331 events) in denosumab users and 51.4% (15,569 events) in bisphosphonate users. The 10-year incidence probability of knee, hip and thumb carpometacarpal OA in denosumab users were 19.2% (95% CI, 18.3%-20.2%; 4,839 events), 14.5% (95% CI, 13.7%-15.3%; 3,396 events), and 5.6% (95% CI, 5.1%-6.1%; 1,241 events) while in bisphosphonate users were 21.4% (95% CI, 20.5%-22.4%; 5,389 events), 14.4% (95% CI, 13.6%-15.3%; 3,388 events), and 6.0% (95% CI, 5.5%-6.6%; 1,297 events). Compared with bisphosphonate, denosumab use was associated with a

reduced risk of any OA (HR 0.96, 95% CI 0.94 to 0.98) and knee OA (HR 0.87, 95% CI 0.84 to 0.91). Use of denosumab was associated with a lower cumulative incidence probability of any OA and knee OA (Figure 2). No significant associations were observed between denosumab use and the incidence of hip or thumb carpometacarpal OA subtypes (Table 2, Supplementary Table 2-5).

Subgroup analysis

To examine whether individuals with certain characteristics might benefit more from denosumab than from bisphosphonate, we performed subgroup analysis stratified by sex, race (White, Black or African American, and Asian), age (45-65 years versus 65 years or older), and BMI (<25, 25-30, and ≥ 30). In participants with an Asian ethnic background, denosumab use was significantly associated with reduced incidences of any OA (HR 0.77, 95% CI 0.70 to 0.84, P for interaction < 0.0001), and knee OA (HR 0.73, 95% CI 0.63 to 0.84, P for interaction = 0.005), while in participants without an Asian ethnic background, a significant association was observed only between denosumab use and knee OA in Caucasian (HR 0.93, 95% CI 0.89 to 0.97). Denosumab use was significantly associated with reduced incidences of knee OA in females (HR 0.87, 95% CI, 0.83 to 0.91) but not in males (HR 1.04, 95% CI, 0.88 to 1.23). Similarly, significant associations between denosumab use and incident OA were only observed in participants over 65 years old (HR 0.95, 95% CI 0.92 to 0.97 for any OA; HR 0.88, 95% CI 0.84 to 0.91 for knee OA) but not in those 45-65 years old. No significant interactions were observed between BMI and denosumab use in the risk of OA (Figure 3).

Sensitivity analysis

In the sensitivity analyses, the lower risks of any OA and knee OA associated with denosumab use remained generally consistent when splitting injectable and oral bisphosphonate products (Supplementary Table 6). A significantly lower risk of knee OA associated with denosumab use remained consistent across varying lag times for the start of follow-up. However, no statistically significant associations were found between denosumab use and any OA diagnosis when considering various lag times for the start of follow-up (Supplementary Table 7). Sensitivity analysis by excluding patients who initiated denosumab and subsequently received bisphosphonates also exerted similar findings, supporting the robustness of primary results (Supplementary Table 8).

Discussion

In this propensity score-matched real-world study with a 10-year follow-up period, among participants diagnosed with osteoporosis, denosumab use was associated with a 13%-27% reduced risk of incident any OA and knee OA, compared with those using bisphosphonates. Older (≥ 65 years) female Asian participants may benefit more from denosumab use than their counterparts in OA risk compared with those using bisphosphonate.

Denosumab has been widely used to treat osteoporosis, but its potential for managing OA remains unclear ¹¹. Previous experimental studies found that denosumab could attenuate OA by protecting cartilage and modulating subchondral bone remodeling ⁶. Recently, a phase 2 randomized placebo-controlled clinical trial found that denosumab had structure-modifying effects in erosive hand OA at the primary endpoint of week 24, with even stronger effects observed at week 48, and no more adverse events compared with placebo ⁸. Another randomized placebo-controlled clinical trial revealed that denosumab could prevent early perioperative bone loss after uncemented total hip arthroplasty ¹². Our current study is the first to investigate the association between denosumab use and incident hip and knee OA. The active comparator bisphosphonate was prescribed for the same indication and patient profile, improving clinical comparability versus non-users. Although differences in administration reliability between denosumab and oral bisphosphonates could introduce adherence bias, our sensitivity analysis using only injectable bisphosphonates yielded consistent results, suggesting minimal impact on the findings. Moreover, differences in baseline risk are smaller and more measurable ¹³. A recent population-based cohort study found that denosumab use was associated with a lower risk of incident type 2 diabetes compared with bisphosphonate use in adults with osteoporosis ¹⁴. Using a large real-world database and a sophisticated study design, we found that denosumab use, compared with bisphosphonate, was significantly associated with a reduced risk of knee OA but not hip OA. The disparities between knee and hip OA may be due to different etiologies ^{15, 16}. Further clinical trials are needed to verify the current findings. In addition, while denosumab was associated with a statistically significant 4% absolute risk reduction for any OA, the clinical significance of this modest effect in a large-scale study should be interpreted with caution.

Our study also found that people with an Asian ethnic background might benefit more from denosumab use than from bisphosphonate, compared with White or African American. However, this finding should be interpreted with caution due to the relatively small sample size of the Asian subgroup (n=4,734), which represented approximately 8% of the total study population. Although individuals of Asian descent generally had lower average bone mineral density (BMD) than those of Caucasian or African descent¹⁷, there were no documented differences in the effectiveness of increasing BMD for denosumab across different racial and ethnic groups¹⁸. The effect of denosumab on bone turnover also appears to be consistent across different ethnicities¹⁹. On one hand, differences in diet, lifestyle, socioeconomic factors, access to healthcare, and prevalence of other health conditions across ethnic groups can influence the observed different effects of denosumab^{20,21}, making it difficult to isolate ethnicity as a single factor of effectiveness. On the other hand, genetic variations affecting bone metabolism and cartilage health might differ across ethnic groups^{22,23}, potentially influencing an individual's response to denosumab.

Both OA and osteoporosis are major global health problems with a high prevalence, imposing significant burdens of morbidity and disability^{24,25}. Moreover, up to 36% of OA patients also have osteoporosis²⁶. Our previous observational studies have shed light on a commonality: heightened bone loss occurs not only in osteoporosis but also in the early stages of OA²⁷. Meanwhile, inflammation has emerged as a shared risk factor for both OA and osteoporosis^{28,29}. Healthcare providers must consider the overlap, especially in older adults and females. Thus, combined screening and management strategies can be implemented. Preclinical studies in animal models of OA have demonstrated that bisphosphate therapy attenuates subchondral bone turnover and bone loss while preserving cartilage volume⁶. The current study provides population-level and real-world evidence that denosumab use for osteoporosis in adults may simultaneously reduce the risk of knee OA, particularly in females, older adults, and individuals of Asian ethnicity. These findings have important implications for tailoring individualized drug management of knee OA.

This study also has limitations. First, the predominance of non-Hispanic White participants with relatively higher socioeconomic status and healthcare access may limit generalizability, and the intensive medical surveillance inherent to an osteoporosis cohort may have increased OA detection compared with the general

population, in whom OA is often underdiagnosed. Second, although we used a new user, active comparator cohort, and a PSM strategy to minimize confounding by indication, residual confounding bias (e.g. causes of osteoporosis and indication bias) remains possible in this observational study. Third, although continuous users of bisphosphonates represent the most widely used drug pattern, other drug sequences (romosozumab, teriparatide, selective estrogen receptor modulator) can also be used in specific clinical scenarios, and thus could be examined as active comparators in future studies. Fourth, because longitudinal weight data were inconsistently captured, we could not assess weight change over time, leaving a potential residual confounder. Last, due to the nature of observational study design, the causality of the results should be interpreted with caution. As no randomized clinical trials have been performed to examine the preventive effect of denosumab on incident knee or hip OA, this study can be viewed as hypothesis-generating and an incentive for future randomized controlled trials to be performed.

Conclusion

In this real-world study, denosumab use was associated with a reduced risk of incident knee OA compared with bisphosphonates in adults with osteoporosis, with the association being more pronounced among older adults, females, and individuals of Asian ethnicity. This study suggests that denosumab could attenuate the development of knee OA, warranting further clinical trials.

Table 1. Baseline characteristics of participants

	Before PSM			After PSM		
	Denosumab	Bisphosphonate	SMD	Denosumab	Bisphosphonate	SMD
Number of participants	59228	161044		59157	59157	
Age at index date	69.6±8.3	67.5±8.2	0.2491	69.6±8.3	69.5±8.1	0.0118
Sex						
Female	51921(87.7%)	141026(87.6%)	0.0028	51875(87.7%)	52125(88.1%)	0.0130
Male	4312(7.3%)	13787(8.6%)	0.0474	4305(7.3%)	4067(6.9%)	0.0157
Ethnicity						
Not Hispanic or Latino	43913(74.1%)	117524(73.0%)	0.0264	43865(74.2%)	43944(74.3%)	0.0031

Hispanic or Latino	1875(3.2%)	9246(5.7%)	0.125 1	1875(3.2%)	1880(3.2%)	0.000 5
Race						
White	44221(74.7 %)	116919(72.6 %)	0.046 8	44179(74.7 %)	44569(75.3%)	0.015 2
Black or African American	1897(3.2%)	9996(6.2%)	0.142 2	1897(3.2%)	1871(3.2%)	0.002 5
Asian	4899(8.3%)	11614(7.2%)	0.039 7	4893(8.3%)	4720(8.0%)	0.010 7
Socioeconomic and psychosocial circumstances	253(0.4%)	1053(0.7%)	0.030 9	253(0.4%)	244(0.4%)	0.002 4
BMI	24.9±5.3	25.4±5.4	0.097 0	24.9±5.3	24.9±5.2	0.009 8
eGFR – CKD-EPI Equations	76.5±17.4	79.0±16.7	0.145 4	76.5±17.4	77.1±16.3	0.037 5
Lifestyle						
Nicotine dependence	1677(2.8%)	7800(4.8%)	0.104 9	1677(2.8%)	1551(2.6%)	0.013 1
Alcohol related disorders	305(0.5%)	1753(1.1%)	0.064 3	305(0.5%)	264(0.4%)	0.010 0
Baseline healthcare encounter						
Preventive medicine services	3650(6.2%)	17470(10.8%)	0.168 6	3650(6.2%)	3694(6.2%)	0.003 1
Inpatient encounter	4515(7.6%)	15364(9.5%)	0.068 5	4514(7.6%)	4339(7.3%)	0.011 2
Baseline co-morbidity						
Hypertensive diseases	16375(27.6 %)	54820(34.0%)	0.138 8	16367(27.7 %)	15768(26.7%)	0.022 8
Ischemic heart diseases	3084(5.2%)	9155(5.7%)	0.021 1	3080(5.2%)	2854(4.8%)	0.017 5
Heart failure	1221(2.1%)	3677(2.3%)	0.015 2	1216(2.1%)	1126(1.9%)	0.010 9
Diabetes mellitus	4251(7.2%)	16149(10.0%)	0.101 8	4249(7.2%)	4105(6.9%)	0.009 5
Dyslipidemia	17601(29.7 %)	58058(36.1%)	0.135 1	17595(29.7 %)	17023(28.8%)	0.021 3
Acute kidney failure	465(0.8%)	1813(1.1%)	0.035 0	465(0.8%)	462(0.8%)	0.000 6
Chronic kidney disease	2423(4.1%)	5347(3.3%)	0.040 8	2413(4.1%)	2334(3.9%)	0.006 8

Diseases of liver	1465(2.5%)	4734(2.9%)	0.028 7	1463(2.5%)	1400(2.4%)	0.006 9
Spondylopathies	4066(6.9%)	9960(6.2%)	0.027 6	4049(6.8%)	3850(6.5%)	0.013 5
Inflammatory polyarthropathies	2331(3.9%)	5886(3.7%)	0.014 7	2327(3.9%)	2187(3.7%)	0.012 4
Depressive episode	3674(6.2%)	12508(7.8%)	0.061 4	3673(6.2%)	3507(5.9%)	0.011 8
Chronic lower respiratory diseases	5610(9.5%)	18123(11.3%)	0.058 5	5602(9.5%)	5079(8.6%)	0.030 9
Gout	261(0.4%)	830(0.5%)	0.010 8	259(0.4%)	233(0.4%)	0.006 8
Neoplasms	13908(23.5%)	29383(18.2%)	0.129 1	13862(23.4%)	13458(22.8%)	0.016 2
Malignant neoplasm of bone	69(0.1%)	78(0.0%)	0.023 7	66(0.1%)	36(0.1%)	0.017 3
Co-medication						
Corticosteroids for systemic use	11905(20.1%)	33714(20.9%)	0.020 7	11887(20.1%)	11173(18.9%)	0.030 5
NSAIDs	7118(12.0%)	23069(14.3%)	0.068 2	7109(12.0%)	6794(11.5%)	0.016 5
Opioids	8648(14.6%)	24607(15.3%)	0.019 0	8633(14.6%)	8170(13.8%)	0.022 4
Estrogens	2410(4.1%)	6947(4.3%)	0.012 2	2406(4.1%)	2317(3.9%)	0.007 7
Selective estrogen receptor modulators	458(0.8%)	957(0.6%)	0.021 7	456(0.8%)	470(0.8%)	0.002 7
Teriparatide	714(1.2%)	737(0.5%)	0.082 4	681(1.2%)	623(1.1%)	0.009 4
Bone mass measurement						
DXA Lumbar spine [Z-score]	92(0.2%)	668(0.4%)	0.048 7	92(0.2%)	230(0.4%)	0.044 8
DXA Lumbar spine [T-score]	53(0.1%)	441(0.3%)	0.043 3	53(0.1%)	161(0.3%)	0.043 0

The baseline period was defined as the time interval within 6 months before the index date.

Propensity score matching was based on demographics (age, sex, race, ethnicity, and socioeconomic status), lifestyle, medical utilization and co-morbidities.

Abbreviations: PSM, Propensity Score Matching; SMD, standardized mean difference; BMI, body mass index; NSAIDs, Nonsteroidal Antiinflammatory Drugs; DXA, dual energy x-ray absorptiometry.

Table 2. The cumulative probability and hazard ratio for the risk of osteoarthritis after the index date among propensity score-matched cohorts

	1-year				Event s	5-year			Event s	10-year		
	Eve nts	Cumulati ve	Crude HR	Adjusted HR		Cumulati ve	Crude H R	Adjusted HR		Cumulati ve	Crude H R	Adjusted HR
		probabilit y	(95% CI)	(95% CI)*		probabilit y	(95% CI)	(95% CI)*		probabilit y	(95% CI)	(95% CI)*
	(95% CI)					(95% CI)				(95% CI)		
Any												
Osteoarthritis												
Denosumab (n= 59157)	532 5	9.4 (9.1- 9.6)	1.08 (0.9 9-1.18)	0.90 (0.8 7-0.94)	1275 3	32.0 (31.6- 32.5)	1.03 (1.0 1-1.05)	0.97 (0.95- 1.00)	1533 1	50.8 (49.6- 52.0)	1.04 (1.02- 1.05)	0.96 (0.94- 0.98)
Bisphosphonat e (n=59157)	582 7	10.4 (10.1- 10.7)	Referenc e	Referenc e	1279 7	32.9 (32.4- 33.4)	Referenc e	Referenc e	1556 9	51.4 (50.2- 52.5)	Referenc e	Referenc e
Knee												
osteoarthritis												
Denosumab (n= 59157)	130 3	2.3 (2.2- 2.4)	0.91 (0.86- 0.97)	0.82 (0.76- 0.88)	4025	10.1 (9.8- 10.4)	0.95 (0.92- 0.98)	0.89 (0.85- 0.92)	4839	19.2 (18.3- 20.2)	0.95 (0.92- 0.98)	0.87 (0.84- 0.91)
Bisphosphonat e (n=59157)	156 9	2.8 (2.7- 3.0)	Referenc e	Referenc e	4434	11.4 (11.1- 11.8)	Referenc e	Referenc e	5389	21.4 (20.5- 22.4)	Referenc e	Referenc e
Hip												
osteoarthritis												
Denosumab (n= 59157)	793	1.4 (1.3- 1.5)	1.02 (0.94- 1.10)	0.87 (0.79- 0.96)	2830	7.1 (6.8- 7.4)	1.09 (1.0 5-1.14)	1.00 (0.95- 1.05)	3396	14.5 (13.7- 15.3)	1.10 (1.0 6-1.14)	0.98 (0.9 4-1.03)
Bisphosphonat e (n=59157)	897	1.6 (1.5- 1.7)	Referenc e	Referenc e	2762	7.1 (6.8- 7.3)	Referenc e	Referenc e	3388	14.4 (13.6- 15.3)	Referenc e	Referenc e
Thumb												
carpometacarpal												
osteoarthritis												
Denosumab (n= 59157)	283	0.5 (0.4- 0.5)	0.91 (0.79- 1.04)	0.83 (0.71- 0.98)	1036	2.6 (2.4- 2.8)	1.03 (0.96- 1.10)	0.96 (0.88- 1.05)	1241	5.6 (5.1- 6.1)	1.05 (0.9 9-1.11)	0.94 (0.87- 1.01)
Bisphosphonat e (n=59157)	336	0.6 (0.5- 0.6)	Referenc e	Referenc e	1050	2.7 (2.5- 2.9)	Referenc e	Referenc e	1297	6.0 (5.5- 6.6)	Referenc e	Referenc e

Bond denotes statistically significant.

Abbreviations: HR, Hazard ratio; CI, confidence interval

Adjusted for demographic characteristics, lifestyle factors, healthcare utilization, comorbid conditions, and concomitant medications.

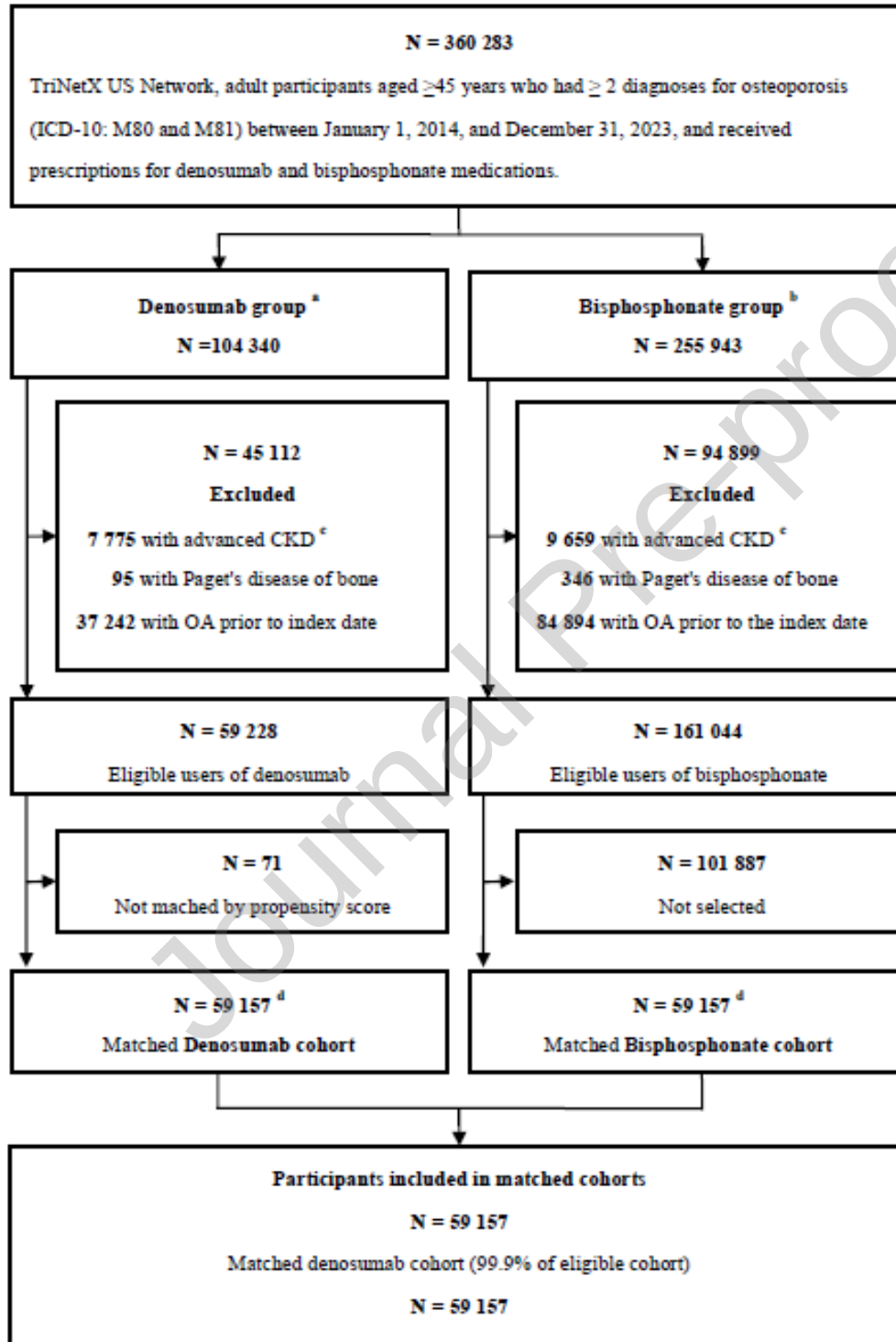


Figure 1. Flowchart for sample selection. a: individuals who received denosumab treatment after their osteoporosis diagnosis. b: individuals who received bisphosphonate treatment after their osteoporosis diagnosis. c: advanced chronic kidney disease defined as eGFR <30 mL/min/1.73 m², dialysis, or an end-

stage renal disease diagnosis with ICD10 codes N18.4-N18.6. d: 1:1 Propensity score matching was based on demographics (age, sex, race, ethnicity, and socioeconomic status), lifestyle, health care utilization, comorbidities, and co-medication.

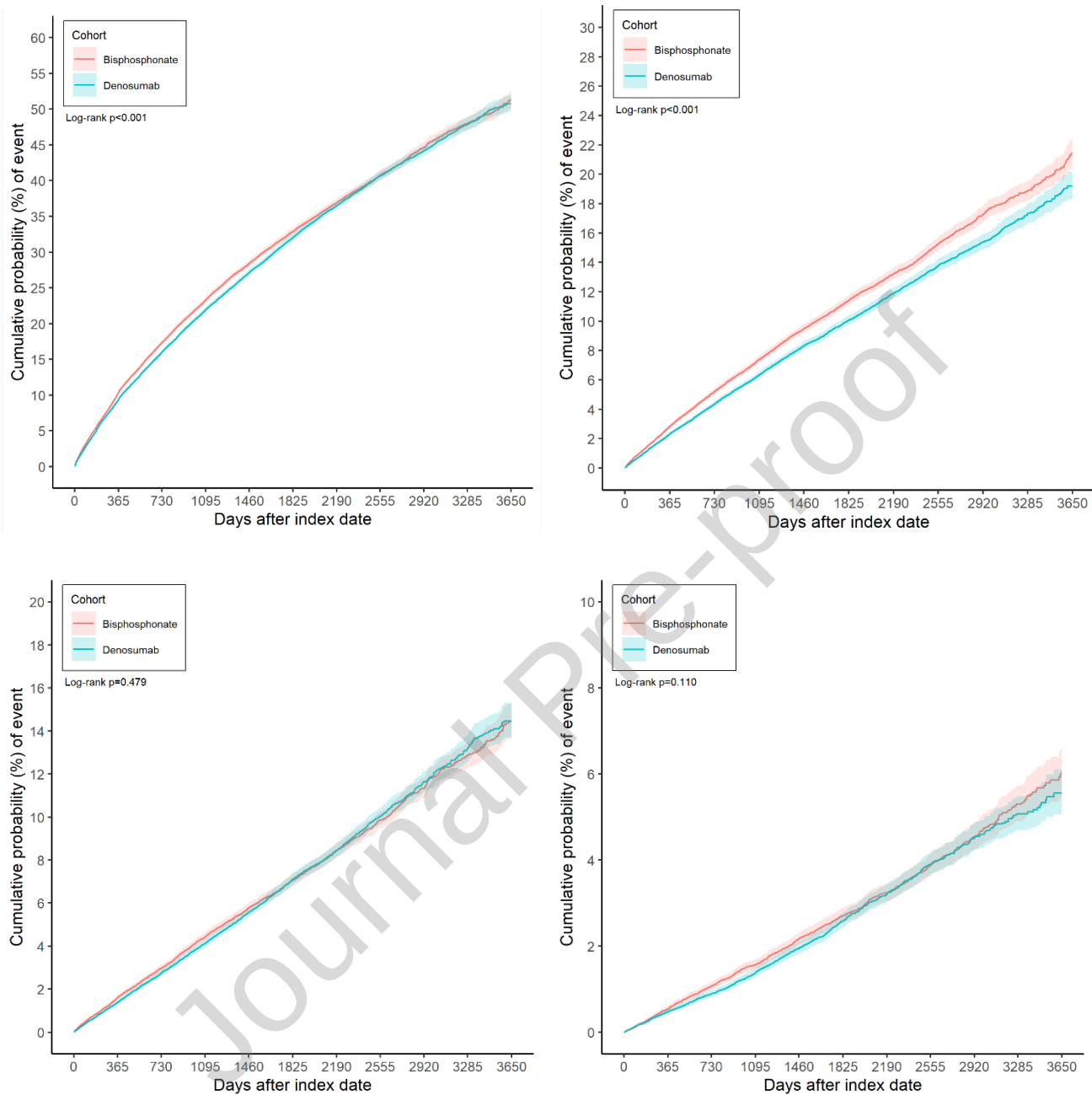
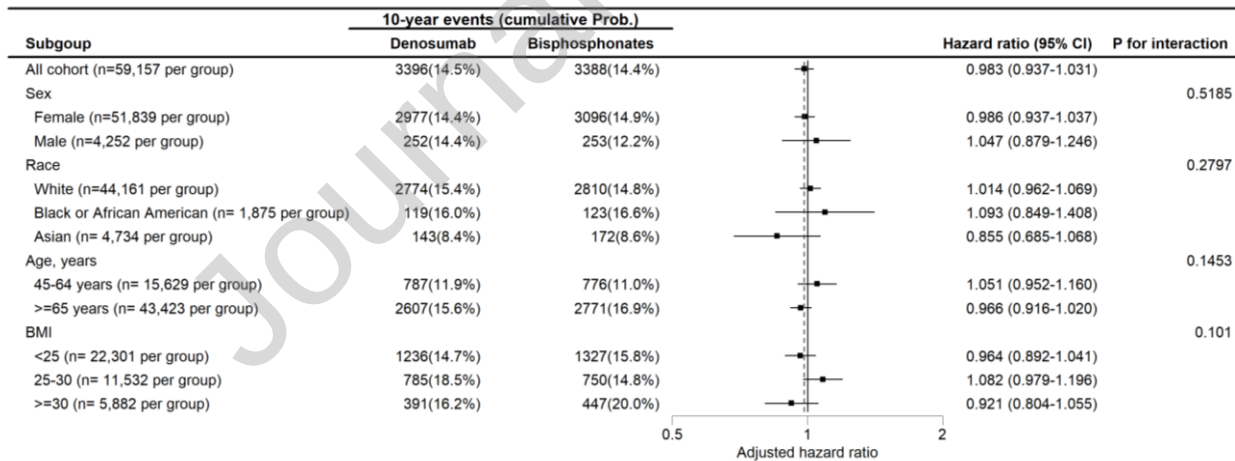
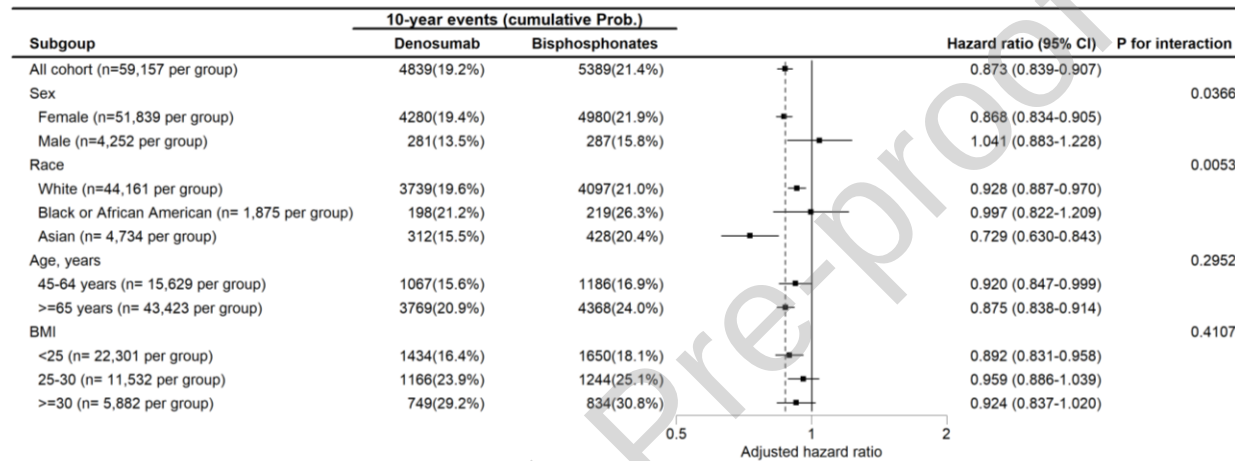
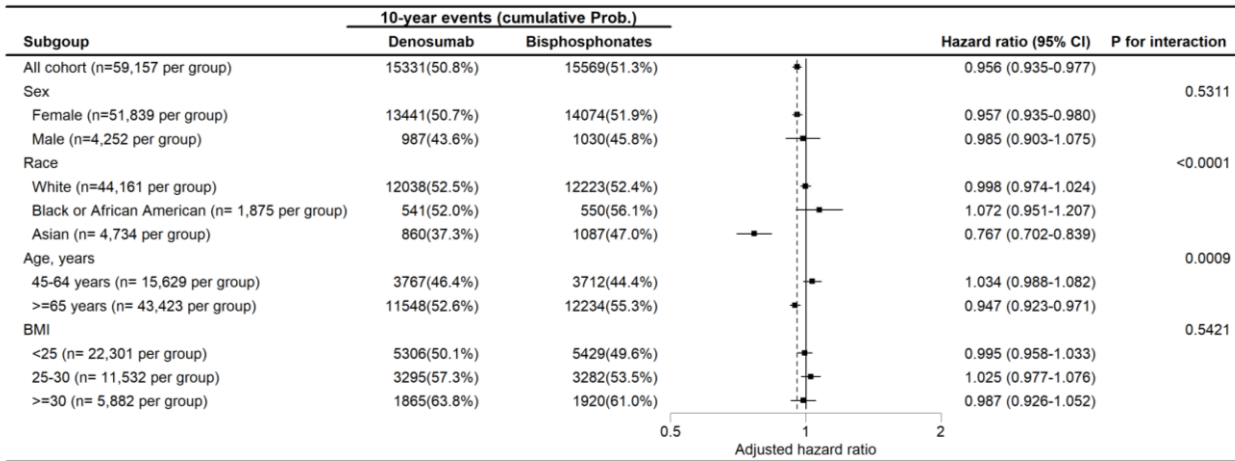


Figure 2. The cumulative risk of osteoarthritis between denosumab and bisphosphonate users. Any Osteoarthritis, (B) Knee Osteoarthritis, (C) Hip Osteoarthritis, (D) Thumb carpometacarpal Osteoarthritis.



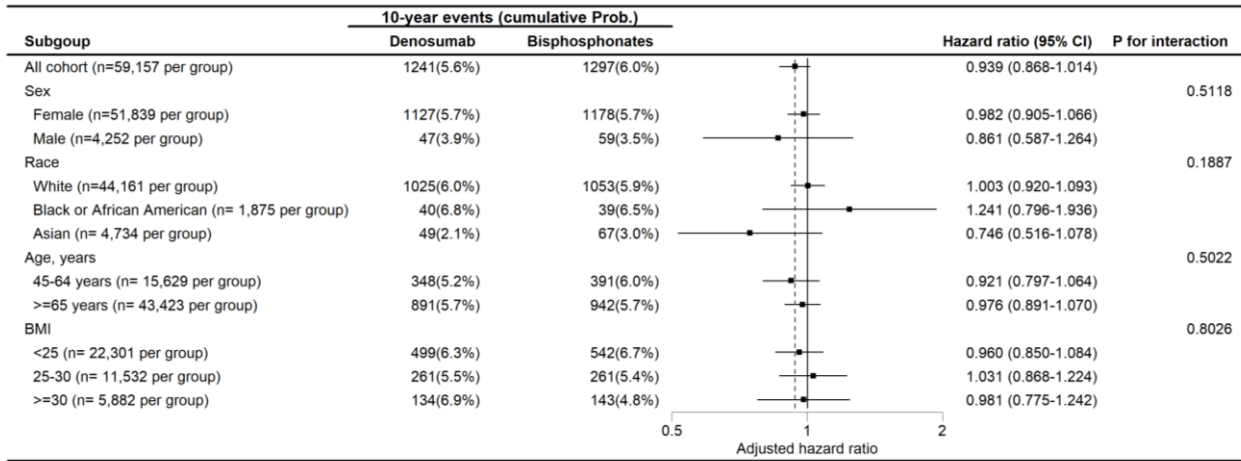


Figure 3. Subgroup analysis for risk of osteoarthritis in participants used denosumab compared with bisphosphonate stratified by age, sex, race, and body mass index. Any Osteoarthritis, (B) Knee Osteoarthritis, (C) Hip Osteoarthritis, (D) Thumb carpometacarpal Osteoarthritis.

Ethical approval

The use of TriNetX in this research was approved by the Institutional Review Board of Chung Shan Medical University Hospital (CSMUH), under the approval number CS2-21176.

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Author contribution

ZZ, JH, CD, JW, and DH were involved in designing the research, extracting data from the website, and directing the study. ZZ, WW and JH took on the tasks of writing the manuscript and conducting the analyses. JW, CD and JH was responsible for accessing and verifying the data reported within the study. JH, WW, SL, HZ, QW, KF, and ZZ focused on developing analytic skills and interpreting the results. WW, SL, HZ, JH, and QW dealt with processing the figures. All authors reviewed and revised the drafted manuscript and gave their approval to the final version. All authors had complete access to all the data in the study and accepted the responsibility of submitting it for publication.

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Conflicts of interest disclosure

This study did not involve any research conducted on human or animal participants. Dr Hunter reported providing consulting advice on scientific advisory boards for Pfizer, Lilly, Haleon, TLCBio, Novartis, Tissuegene, Biobone, Sanofi, and Enlivex outside the submitted work. No other disclosures were reported.

Data availability statement

The data analyzed in this study are subject to the following licenses/restrictions: Data in this study were retrieved from TriNetX US Network. All data available in the database was administered by the TriNetX platform. Detailed information can be retrieved at the official website of the research network (<https://trinetx.com>). Requests to access these datasets should be directed to the TriNetX platform (<https://trinetx.com>).

Consent to participate

Not applicable.

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